

IoT-ENABLED AIR POLLUTION METER

With Digital Dashboard On Smartphone

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Presented here is a project to monitor air quality on your smartphone using Blynk application and Arduino board. Blynk is an Internet of Things (IoT) platform to control Arduino, Raspberry Pi and the like over the Internet. In this project Blynk provides a digital dashboard on your smartphone that displays real-time air quality readings for the immediate surroundings.

Blynk is not meant for a specific board or shield. It will get you online and ready for the IoT, irrespective of whether Arduino or Raspberry Pi is linked to the Internet over Wi-Fi, Ethernet or an ESP8266 chip.

How Blynk works

Blynk can control hardware remotely. It can display sensor data, store and visualise it, among other cool things. There are three major components in the platform as given below:

Blynk app. It allows you to create amazing interfaces for your projects using various widgets.

Blynk server. It is responsible for all communication between the smartphone and hardware. You can use Blynk Cloud or run your private Blynk server locally. It is open source, and can handle thousands of devices. It can also be launched on Raspberry Pi.

Blynk libraries. Libraries are available for all popular hardware platforms. The libraries enable

communication with the server, and process all incoming and outgoing commands.

When you press a button on Blynk app on the phone, the signal travels to the Blynk Cloud, where it finds its way to the hardware. It works the same way in the opposite direction, and everything happens in a blink of an eye. The concept is shown in Fig. 1.

Air pollution meter

The following hardware components are used in the air pollution meter:

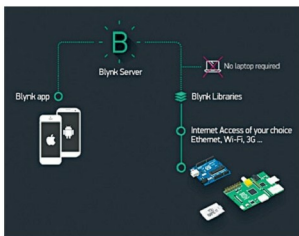


Fig. 1: Blynk gets connected with embedded hardware

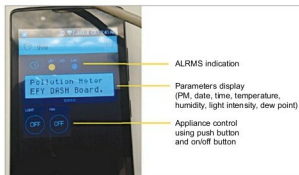


Fig. 2: Digital dashboard on Android phone

Controller and sensors.

Arduino Uno is used as a control and measuring unit, and Arduino Uno Ethernet shield for connection to the Blynk Cloud. The sensors include nova PM sensor SDS011, gas sensor MQ135, and temperature and humidity sensor DHT11.

Smartphone and digital dashboard.

Android phone is used as a smartmeter. Blynk app is a well-designed interface builder. It works on iOS and Android, and provides easy-to-use widgets for the LCD, push buttons, on/off buttons, LED indication, RTC and more. Using Blynk digital dashboard features in a smartphone helps reduce the cost of the project because traditional LCD display, mechanical buttons, LEDs, RTC and so on are not required. The digital dashboard or smartmeter display is shown in Fig. 2.

Circuit and working

Circuit diagram of the IoT-air pollution meter with digital dashboard is shown in Fig. 3. Heart of the circuit is Arduino Uno board with Arduino shield. Other components used are voltage regulators 7805 (IC1 and IC2), temperature and humidity module DHT11 connected to connector CON3, gas sensor MQ135 connected to connector CON2, PM2.5/PM10 sensor connected to connector CON1 and a few others.

PM2.5/PM10 sensor (SDS011).

Particle pollution, also called particulate matter (or PM), is a mixture